

Hybrid Feature-Based Detection of AI-Generated Text

Finn Fonteijn

s2899078

f.f.fonteijn@student.utwente.nl

Linn Gregussen

s3422372

l.y.gregussen@student.utwente.nl

Abstract

1 Introduction

For this project, we explore which linguistic or statistical features are most effective for distinguishing human- vs. AI-generated text. Recent surveys indicate that hybrid approaches, combining statistical and neural methods, often outperform single-modality detectors in robustness and generalization (Wu et al., 2025; Su and Wu, 2024).

The project approach is two-fold. First, a feature analysis is conducted to understand which classic linguistic and statistical features, e.g. sentence complexity, punctuation patterns, lexical diversity, perplexity, and part-of-speech (POS) distributions, are most discriminative. Multiple lightweight classifiers (logistic regression, random forest, XGBoost) are trained on the extracted linguistic/statistical features and their feature importance rankings are compared to identify important/robust signals. Second, the project explores how to make an AI-detector harder to fool in a real world-deployment scenario, specifically, increasing its robustness against adversarial attacks and domain shifts. This is done by combining multiple classification approaches (including neural methods like BERT and statistical methods like Naive Bayes) via majority voting.

RQ: Which linguistic or statistical features are most effective in distinguishing human- vs. AI-generated text?

2 Related Work

3 Data

To ensure our model generalizes across diverse English language sources, domains, and LLMs, and to mitigate overfitting to any single dataset’s biases, we will utilize multiple datasets for training and evaluation. This approach allows for robust testing of features and hybrids in varied contexts, such as essays, Q&A, and long-form texts from different LLMs.

Dataset	Samples	Human (%)	AI (%)
HC3	85K	68.5	31.5
sunilthite	29K	60.1	39.9
DAIGT V2	45K	61.0	39.0
AH&AITD	12K	50.0	50.0

Table 1: Dataset size and class distributions

- AH&AITD (Arslan’s Human and AI Text Database): texts across domains (articles, abstracts, stories, news, reviews) (Akram, 2023).
- LLM - Detect AI Generated Text Dataset (Kaggle sunilthite): essays (Thite, 2023).
- DAIGT V2 Train Dataset (Kaggle): essays (Deotte, 2023).
- Human ChatGPT Comparison Corpus (HC3): Q&A from domains (finance, medicine, law, open) (Guo et al., 2023).

3.1 Dataset Balancing

The datasets show varying degrees of class imbalance between the target classes human-written (0) and AI-generated (1). With the extremes being the LLM Detect Competition being 99.8% human-written, while AH&AITD is perfectly balanced. The dataset’s class distribution is shown in Table 1.

The dataset was balanced using a per-source strategy, where each source contributes 50% human and 50% AI-generated samples according to its configured proportion of the total dataset size.

4 Method

The project methodology consists of two efforts to address different aspects of AI text detection: a feature-based analysis and an ensemble-based detector.

4.1 Approach 1: Feature-Based Analysis

The feature-based analysis focuses on understanding which linguistic and statistical features are most informative in classifying human- vs. AI-generated text. For each text in the dataset, a set of features are extracted (section 4.1.1). These features are then used to train three lightweight classifiers: logistic regression and random forest using scikit-

learn (Pedregosa et al., 2011), and XGBoost (Chen and Guestrin, 2016).

The dataset was balanced as described in section 3.1, and an 80/20 train-test split was used for all models. Feature importance rankings were retrieved and compared across these three models, giving insights on which linguistic patterns are consistently important for distinguishing human- vs. AI-generated text.

4.1.1 Feature Extraction

For each entry in the dataset, the following features were extracted using the spaCy library (Honninger et al., 2020) with the en_core_web_sm model for all linguistic processing tasks, including tokenization, lemmatization, part-of-speech tagging, and dependency parsing.

Lexical Features Four lexical features were chosen and extracted. Average word length and word length variance to measure vocabulary complexity. Root Type-Token Ratio (RTTR), and hapax legomena ratio to measure lexical diversity.

The Root Type-Token Ratio (RTTR) is a token-type ratio that accounts for document size by normalizing by the square of the document length (Reviriego et al., 2024). It was calculated by lemmatizing the text, counting the unique types and dividing by squared text length. A hapax legomenon refers to a word (or expression) occurring only once in a context, and here measures the proportion of words that appear exactly once in a sample.

Syntactic Features Sentence-level statistics were chosen to measure sentence complexity, and include average sentence length and variance. Parse tree depth features (average, maximum and variance) were extracted to measure syntactic complexity. Parse depth is computed as the maximum number of ancestors for any token in a sentence.

Punctuation Features The ratio of punctuation marks to word count was calculated for comma, period, exclamation mark, and question mark. Additionally, the overall punctuation ratio was included.

Part-of-Speech Features POS tags were extracted using spaCy's coarse-grained POS tags (token.pos_) rather than fine-grained tags (token.tag_) to capture overall patterns in word class usage, fine grained tags were excluded as they may be too specific for the objective, and yield sparse data. The ratio of each POS category (noun, verb, adjective, adverb, etc.) to total tokens

was calculated as well as a noun-to-verb ratio as a derived feature.

Perplexity Perplexity is defined as the inverse test set probability according to the model used for calculating the perplexity score, normalized by the length of the text (Jurafsky and Martin, 2025). It is an evaluation of the model's ability to predict uniformly among the set of tokens in a corpus, or in other words how "surprised" the model is by the text, where a lower perplexity indicates more predictable/less surprising text for the model. Perplexity was calculated using EleutherAI's Pythia 1.4B model (Biderman et al., 2023), an open-source model trained on the Pile dataset (Gao et al., 2020). The model was downloaded from Hugging Face, and run using the transformers library (Wolf et al., 2020). This model was chosen for its balance between computational efficiency and generalization due to its diverse training data, which includes web text, books, and academic sources similar to the datasets of this experiment.

Burstiness Features

4.1.2 Feature Selection

To reduce collinearity, a correlation analysis was performed on all the described extracted features, and from pairs with high pairwise correlation ($r > 0.8$), one feature was dropped. The feature to be dropped was selected based on whether it was correlated with more other features, and which feature would be more informative in the project context. The full list of selected and dropped features is shown in Table 2.

4.2 Approach 2: Ensemble-Based Detector

The ensemble-based detector aims to build a practical, robust detector by combining multiple classifiers via majority voting. The ensemble includes: a BERT classifier; a Naive Bayes classifier; potentially a feature-based model (TODO update after discussion!!). The ensemble works by letting each model making a prediction, then "voting", selecting the majority class. This approach increases robustness against adversarial attacks and domain shifts, as fooling the ensemble would require deceiving a majority of its components simultaneously.

4.3 Evaluation

Classification Performance: All classifiers are evaluated using accuracy, precision, recall, F1-score, and confusion matrices on held-out test sets,

Table 2: List of selected/considered linguistic features

Category	Feature Name
Lexical	average_word_length
	word_length_variance
	rtrr_root_type_token_ratio
	hapax_legomena_ratio
Syntactic	average_sentence_length
	sentence_length_variance
	average_parse_depth
	maximum_parse_depth
	parse_depth_variance
Punctuation	comma_ratio
	period_ratio
	exclamation_mark_ratio
	question_mark_ratio
	punctuation_per_word
Part-of-Speech (POS)	pos_ratio_<tag>
	noun_to_verb_ratio
Burstiness	sentence_length_burstiness
	sentence_length_iqr
	position_burstiness_diff
	long_sentence_ratio
	word_length_burstiness
	clause_length_burstiness
	sentence_length_cv [†]
	sentence_length_range_ratio [†]
	adjacent_sentence_variation [†]
Other	short_sentence_ratio [†]
	perplexity

[†] Removed; high correlation ($r > 0.8$) with other feature(s)

for the ensemble both the individual models and the complete ensemble performance is reported.

Feature Importance Analysis: For the feature-based analysis, we extract and compare feature importances from each model: absolute coefficient values from logistic regression (normalized to sum to 1), Gini importance from random forest, and gain-based importance from XGBoost. By ranking features across models, we identify which linguistic patterns are consistently discriminative regardless of the classification algorithm.

The feature importance analysis compares three classifiers to identify robust signals that generalize across different models (linear vs. tree-based, bagging vs. boosting).

5 Experiment and Results

6 Discussion and Conclusion

References

Arslan Akram. 2023. [AH&AITD: Arslan’s Human and AI Text Database](#). Dataset. Associated with the article: An Empirical Study of AI-Generated Text

Detection Tools. *Advances in Machine Learning & Artificial Intelligence*, 4(2), 44–55.

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. 2023. [Pythia: a suite for analyzing large language models across training and scaling](#). In *Proceedings of the 40th International Conference on Machine Learning, ICML’23*. JMLR.org.

Tianqi Chen and Carlos Guestrin. 2016. [Xgboost: A scalable tree boosting system](#). In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD ’16*, page 785–794, New York, NY, USA. Association for Computing Machinery.

Chris Deotte. 2023. [Daigt v2 train dataset](#). Dataset containing 44k human-written essays and AI-generated ones from different LLMs, including OSS ones.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. 2020. [The Pile: An 800gb dataset of diverse text for language modeling](#). *arXiv preprint arXiv:2101.00027*.

Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie, Yuxuan Ding, Jianwei Yue, and Yupeng Wu. 2023. [How close is chatgpt to human experts? comparison corpus, evaluation, and detection](#). *arXiv preprint arxiv:2301.07597*.

Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. [spaCy: Industrial-strength Natural Language Processing in Python](#). Zenodo.

Daniel Jurafsky and James H. Martin. 2025. [Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, with Language Models](#), 3rd edition. Draft. Online manuscript released August 24, 2025.

F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. [Scikit-learn: Machine learning in Python](#). *Journal of Machine Learning Research*, 12:2825–2830.

Pedro Reviriego, Javier Conde, Elena Merino-Gómez, Gonzalo Martínez, and José Alberto Hernández. 2024. [Playing with words: Comparing the vocabulary and lexical diversity of chatgpt and humans](#). *Machine Learning with Applications*, 18:100602.

Yongye Su and Yuqing Wu. 2024. [Robust detection of llm-generated text: A comparative analysis](#). *Preprint*, arXiv:2411.06248.

Sunil Thite. 2023. [Llm - detect ai generated text dataset](#). Dataset containing more than 28,000 essays written by students and AI-generated.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Junchao Wu, Shu Yang, Runzhe Zhan, Yulin Yuan, Lidia Sam Chao, and Derek Fai Wong. 2025. [A survey on llm-generated text detection: Necessity, methods, and future directions](#). *Computational Linguistics*, 51(1):275–338.

A Example appendix